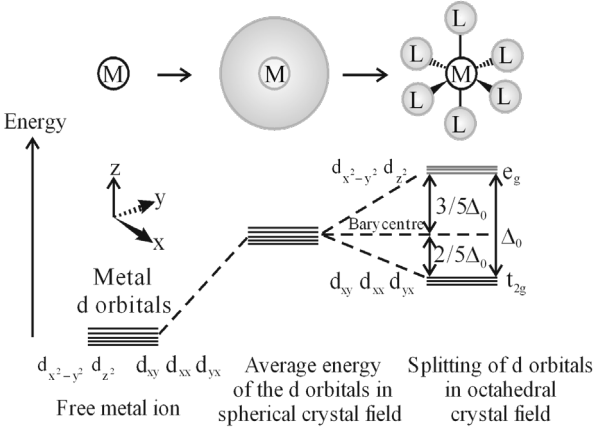
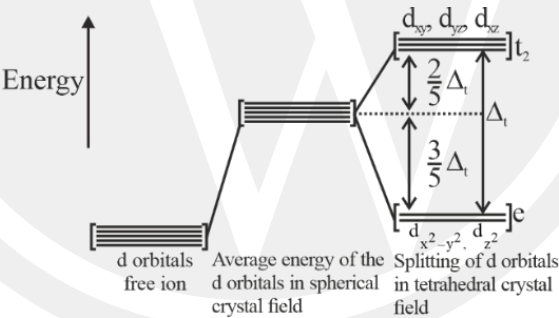
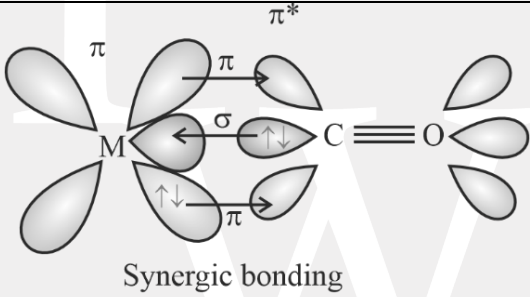


Coordination Compounds

Sr. No.	Concept	Description	Other information																		
1.	Introduction	General formula :- $[M(L)_5]X$	M = Metal, L = Ligand [] = Coordination sphere																		
2.	Werner' Theory	According to Werner most elements exhibit two types of valencies: (a) Primary valency - O.N. (b) Secondary valency - C.N. e.g. $CoCl_3.4NH_3$ Molecular formula $[Co(NH_3)_6]Cl_3$ complex formula H_3N NH_3 Cl H_3N NH_3 Cl Co Werner structure 3 dotted line shows - Primary Valency 6 solid line show - Secondary Valency	O.N. = Oxidation No. C.N. = Co-ordination No.																		
3.	Coordination polyhedrons	L L L L L L M Octahedral L L L M Tetrahedral L L L M Square planar L L L L M Trigonal bipyramidal L L L L M Square pyramidal	M represents the central atom/ion and L, a unidentate ligand																		
4.	Bonding in Coordination Compounds (a) Valence bond theory (i) Magnetic moment	<table><tr><th>Coordination number</th><th>Type of hybridisation</th><th>Distribution of hybrid orbitals in space</th></tr><tr><td>4</td><td>sp^3</td><td>Tetrahedral</td></tr><tr><td>4</td><td>dsp^2</td><td>Square planar</td></tr><tr><td>5</td><td>sp^3d</td><td>Trigonal bipyramidal</td></tr><tr><td>6</td><td>sp^3d^2</td><td>Octahedral</td></tr><tr><td>6</td><td>d^2sp^3</td><td>Octahedral</td></tr></table> $\mu = \sqrt{n(n+2)}B.M$	Coordination number	Type of hybridisation	Distribution of hybrid orbitals in space	4	sp^3	Tetrahedral	4	dsp^2	Square planar	5	sp^3d	Trigonal bipyramidal	6	sp^3d^2	Octahedral	6	d^2sp^3	Octahedral	n is number of unpaired electrons
Coordination number	Type of hybridisation	Distribution of hybrid orbitals in space																			
4	sp^3	Tetrahedral																			
4	dsp^2	Square planar																			
5	sp^3d	Trigonal bipyramidal																			
6	sp^3d^2	Octahedral																			
6	d^2sp^3	Octahedral																			

Sr. No.	Concept	Description	Other information
	(b) Crystal Field Theory	 Octahedral $CFSE = a \times (-0.4\Delta_o) + b \times (0.6\Delta_o) + np$ If $\Delta_o < P$, then complex is high spin Tetrahedral $CFSE = a' \times -(0.6\Delta_t) + b' \times (0.4\Delta_t) + np$ If $\Delta_o > P$, then complex is low spin 	M represents the central atom/ion and L, a unidentate ligand Where Δ_o is octahedral splitting energy and Δ_t is tetrahedral splitting energy a = No. of e^- s in 't_{2g}' orbital. b = No. of e^- s in 'e_g' orbital. a' = No. of e^- s in 'e' orbital. b' = No. of e^- s in 't_2' orbital.
(i)	Spectrochemical series.	$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$	
(ii)	Relation between Δ_t and Δ_o	$\Delta_t = (4/9)\Delta_o$	

Sr. No.	Concept	Description	Other information																																																																																																						
5.	Geometrical and optical isomers	For coordination number 6 <table><tr><th>Formula</th><th>Geometrical Isomers</th><th>Cis</th><th>Trans</th><th>Pairs of enantiomers</th><th>Number of stereoisomers</th></tr><tr><td>Ma₆</td><td>–</td><td>–</td><td>–</td><td>–</td><td>1</td></tr><tr><td>Ma₅b</td><td>–</td><td>–</td><td>–</td><td>–</td><td>1</td></tr><tr><td>Ma₄b₂</td><td>2</td><td>1</td><td>1</td><td>–</td><td>2</td></tr><tr><td>Ma₃b₃</td><td>2</td><td>Fac</td><td>Mer</td><td>–</td><td>2</td></tr><tr><td>Ma₄bc</td><td>2</td><td>1</td><td>1</td><td>–</td><td>2</td></tr><tr><td>Ma₃b₂c</td><td>3</td><td>1</td><td>2</td><td>–</td><td>3</td></tr><tr><td>Ma₃bcd</td><td>4</td><td>1'</td><td>3</td><td>1</td><td>5</td></tr><tr><td>Ma₂b₂c₂</td><td>5</td><td>1'</td><td>4</td><td>1</td><td>6</td></tr><tr><td>Ma₂b₂cd</td><td>6</td><td>2'</td><td>4</td><td>2</td><td>8</td></tr><tr><td>Ma₂bcd₂e</td><td>9</td><td>6'</td><td>3</td><td>6</td><td>15</td></tr><tr><td>Mabcd₂ef</td><td>15'</td><td>–</td><td>–</td><td>15</td><td>30</td></tr></table> For coordination number 4 [Square planar] <table><tr><th>Formula</th><th>Geometrical Isomers</th><th>Cis</th><th>Trans</th><th>Number of stereoisomers</th></tr><tr><td>Ma₄</td><td>–</td><td>–</td><td>–</td><td>1</td></tr><tr><td>Ma₃b</td><td>–</td><td>–</td><td>–</td><td>1</td></tr><tr><td>Ma₂b₂</td><td>2</td><td>1</td><td>1</td><td>2</td></tr><tr><td>Ma₂b₂c</td><td>2</td><td>1</td><td>1</td><td>2</td></tr><tr><td>Mabcd</td><td>3</td><td>–</td><td>–</td><td>3</td></tr></table> CN = 4, Tetrahedral: no geometrical isomer possible, Only optical isomer is possible for Mabcd type structure.	Formula	Geometrical Isomers	Cis	Trans	Pairs of enantiomers	Number of stereoisomers	Ma ₆	–	–	–	–	1	Ma ₅ b	–	–	–	–	1	Ma ₄ b ₂	2	1	1	–	2	Ma ₃ b ₃	2	Fac	Mer	–	2	Ma ₄ bc	2	1	1	–	2	Ma ₃ b ₂ c	3	1	2	–	3	Ma ₃ bcd	4	1'	3	1	5	Ma ₂ b ₂ c ₂	5	1'	4	1	6	Ma ₂ b ₂ cd	6	2'	4	2	8	Ma ₂ bcd ₂ e	9	6'	3	6	15	Mabcd ₂ ef	15'	–	–	15	30	Formula	Geometrical Isomers	Cis	Trans	Number of stereoisomers	Ma ₄	–	–	–	1	Ma ₃ b	–	–	–	1	Ma ₂ b ₂	2	1	1	2	Ma ₂ b ₂ c	2	1	1	2	Mabcd	3	–	–	3	No optical isomerism possible for square planar type complex.
Formula	Geometrical Isomers	Cis	Trans	Pairs of enantiomers	Number of stereoisomers																																																																																																				
Ma ₆	–	–	–	–	1																																																																																																				
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Ma ₄ b ₂	2	1	1	–	2																																																																																																				
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Ma ₄ bc	2	1	1	–	2																																																																																																				
Ma ₃ b ₂ c	3	1	2	–	3																																																																																																				
Ma ₃ bcd	4	1'	3	1	5																																																																																																				
Ma ₂ b ₂ c ₂	5	1'	4	1	6																																																																																																				
Ma ₂ b ₂ cd	6	2'	4	2	8																																																																																																				
Ma ₂ bcd ₂ e	9	6'	3	6	15																																																																																																				
Mabcd ₂ ef	15'	–	–	15	30																																																																																																				
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Ma ₄	–	–	–	1																																																																																																					
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Ma ₂ b ₂ c	2	1	1	2																																																																																																					
Mabcd	3	–	–	3																																																																																																					
6.	Bonding in metal carbonyl	 Synergic bonding	M = metal having vacant d orbital																																																																																																						

Sr. No.	Concept	Description	Other information
7.	Structure of some homolyptic metal complex	$\begin{array}{c} \text{CO} \\ \\ \text{OC} - \text{Ni} - \text{CO} \\ \\ \text{CO} \end{array}$ Ni(CO)_4 Tetrahedral $\begin{array}{c} \text{CO} \\ \\ \text{OC} - \text{Fe} - \text{CO} \\ \\ \text{CO} \end{array}$ Fe(CO)_5 Trigonal bipyramidal $\begin{array}{c} \text{CO} \\ \\ \text{OC} - \text{Cr} - \text{CO} \\ \\ \text{CO} \end{array}$ Cr(CO)_6 Octahedral $\begin{array}{c} \text{CO} \quad \text{CO} \quad \text{CO} \\ \quad \quad \\ \text{OC} - \text{Mn} - \text{Mn} - \text{CO} \\ \quad \quad \\ \text{CO} \quad \text{CO} \quad \text{CO} \end{array}$ $[\text{Mn}_2(\text{CO})_{10}]$ $\begin{array}{c} \text{O} \\ \\ \text{OC} - \text{Co} - \text{C} - \text{Co} - \text{CO} \\ \quad \quad \\ \text{OC} \quad \text{C} \quad \text{CO} \\ \\ \text{O} \end{array}$ $[\text{Co}_2(\text{CO})_8]$	